

Lecture 6 2026-09-17

Last time: equality-constrained N.C.O,
linear least norm

Today: equality-constrained Newton method

Equality-constrained Newton method



$$\min_x f(x) \quad f \in \mathcal{C}^2$$

subject to: $Ax = b$

Solution approach:

1. Write the Lagrangian
2. Derive the N.C.O.
3. Iteratively find solution satisfying N.C.O.

1. Let's call Lagrange multiplier y

$$\mathcal{L}(x, y) = f(x) + y^T (Ax - b)$$

$y^T Ax - y^T b \xrightarrow{\nabla_x = 0} (A^T y)^T x \Rightarrow \nabla_x(\mathcal{L}) = A^T y$

N.C.O.: 1. $\nabla_y \mathcal{L}(x^*, y^*) = 0 = Ax^* - b$

2. $\nabla_x \mathcal{L}(x^*, y^*) = 0 = \nabla_x f(x^*) + A^T y^*$

Define the residual $r(x, y) = \begin{pmatrix} \nabla_x \mathcal{L}(x, y) \\ \nabla_y \mathcal{L}(x, y) \end{pmatrix}$

Solving for $(x^*, y^*) \Rightarrow$ finding where $r(x, y) = 0$
 converts optimization problem
 to root finding for $r(x, y) = 0$

To solve for $r(x^*, y^*) = 0 \rightarrow$ use Taylor series approximation:

$$r(x, y) \approx r(\bar{x}, \bar{y}) + \frac{\partial r}{\partial x} \Delta x + \frac{\partial r}{\partial y} \Delta y = 0$$

$$r(x, y) \approx r(\bar{x}, \bar{y}) + \underbrace{\left. \frac{\partial r}{\partial x} \right|_{\bar{x}, \bar{y}}}_{\text{these are Jacobian matrices}} \Delta x + \underbrace{\left. \frac{\partial r}{\partial y} \right|_{\bar{x}, \bar{y}}}_{\text{these are Jacobian matrices}} \Delta y := 0$$

Solve for $\Delta x, \Delta y$ iteratively

these are Jacobian matrices

Let's write out Jacobians:

$$\begin{aligned} \frac{\partial r}{\partial x} &= \begin{pmatrix} \frac{\partial}{\partial x} (\nabla_x \mathcal{L}(x, y)) \\ \frac{\partial}{\partial x} (\nabla_y \mathcal{L}(x, y)) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x} (\nabla_x f(x) + A^T y) \\ \frac{\partial}{\partial x} (Ax - b) \end{pmatrix} \\ &= \begin{pmatrix} \nabla_{xx}^2 f(x) \\ A \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \frac{\partial r}{\partial y} &= \begin{pmatrix} \frac{\partial}{\partial y} (\nabla_x \mathcal{L}(x, y)) \\ \frac{\partial}{\partial y} (\nabla_y \mathcal{L}(x, y)) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial y} (\nabla_x f(x) + A^T y) \\ \frac{\partial}{\partial y} (Ax - b) \end{pmatrix} \\ &= \begin{pmatrix} A^T \\ 0 \end{pmatrix} \end{aligned}$$

$$\Rightarrow r(x, y) \approx r(\bar{x}, \bar{y}) + \begin{pmatrix} \nabla_{xx}^2 f(\bar{x}) \\ A \end{pmatrix} \Delta x + \begin{pmatrix} A^T \\ 0 \end{pmatrix} \Delta y := 0$$

$$\Rightarrow \begin{pmatrix} \nabla_{xx}^2 f(\bar{x}) \\ A \end{pmatrix} \Delta x + \begin{pmatrix} A^T \\ 0 \end{pmatrix} \Delta y = -r(\bar{x}, \bar{y})$$

$$\rightarrow \underbrace{\begin{pmatrix} \nabla_{xx}^2 f(\bar{x}) & A^T \\ A & 0 \end{pmatrix}}_{\text{Jacobian matrix}} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} = - \begin{pmatrix} \nabla_x f(\bar{x}) + A^T \bar{y} \\ A \bar{x} - b \end{pmatrix}$$

KKT matrix

When is this matrix invertible?

1. A is full rank
2. $\nabla_{xx}^2 f(\bar{x})$ is positive semidefinite

In HW2, we will focus on solving problems satisfying these assumptions:

Convex quadratic program with linear equality constraints

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^T P x + p^T x & P \text{ is positive semidefinite} \\ \text{subj. to: } & Ax = b & A \text{ is full row rank} \end{aligned}$$

Let's evaluate the KKT matrix for this problem:

$$f(x) = \frac{1}{2} x^T P x + p^T x$$

$$\Rightarrow \nabla_x f(x) = Px + p$$

$$\nabla_{xx}^2 f(x) = P \succeq 0$$

P is P.S.D.

$$\Rightarrow \begin{pmatrix} \nabla_{xx}^2 f(\bar{x}) & A^T \\ A & 0 \end{pmatrix} \rightarrow \begin{pmatrix} P & A^T \\ A & 0 \end{pmatrix}$$

$$-r(x, y) = - \begin{pmatrix} \nabla_x f(\bar{x}) + A^T y \\ A\bar{x} - b \end{pmatrix} = - \begin{pmatrix} P\bar{x} + p + A^T y \\ A\bar{x} - b \end{pmatrix}$$

Generalized solution routine:

- Start with $(x^{(0)}, y^{(0)})$
- for $k = 1 \rightarrow N_{\max}$

$K = \text{Construct KKT}(x^{(k-1)}, y^{(k-1)})$

$r = \text{Compute Residual}(x^{(k-1)}, y^{(k-1)})$

$(\Delta x, \Delta y) = \text{Solve KKT}(K, r)$

$\alpha = \text{LineSearch}(\Delta x, \Delta y)$

$$x^k = x^{k-1} + \alpha \Delta x$$

$$y^k = y^{k-1} + \alpha \Delta y$$

if $\|r(x^k, y^k)\|_2 \leq \epsilon_{\text{tol}}$
terminate

KKT matrix:

suppose QP is:

$$\min_x \frac{1}{2} x^T P x + p^T x$$

$$x \in \mathbb{R}^n, P \succeq 0$$

$$\text{subj. to: } Ax = b$$

$$A \in \mathbb{R}^{m \times n}$$

$$\text{KKT: } \begin{pmatrix} P & A^T \\ A & 0 \end{pmatrix} \underbrace{\begin{pmatrix} n \times n & n \times m \\ m \times n & m \times m \end{pmatrix}}_{(m+n) \times (m+n)} \Rightarrow \mathcal{O}((m+n)^3)$$

with the specific form a convex QP, we can
more efficiently invert $(\text{KKT})^{-1}$ using

Feasible start Newton method

Suppose we have $\bar{x}$ such that $A\bar{x} = b$, then an

alternative derivation is:

$$\begin{aligned} & \min_x f(x) \\ & \text{subj. to: } Ax = b \end{aligned}$$

$$\rightarrow f(x) \approx f(\bar{x}) + \nabla_x f(\bar{x})^T \Delta x + \frac{1}{2} \Delta x^T \nabla_{xx}^2 f(\bar{x}) \Delta x$$

$$Ax \approx A(\bar{x} + \Delta x) = \underbrace{A\bar{x}}_{=0} + A\Delta x = A\Delta x = 0$$

$$\begin{aligned} \rightarrow \min_{\Delta x} & f(\bar{x}) + \nabla_x f(\bar{x})^T \Delta x + \frac{1}{2} \Delta x^T \nabla_{xx}^2 f(\bar{x}) \Delta x \\ & \text{subj. to: } A\Delta x = 0 \end{aligned}$$

i.e., we will take steps along $\Delta x \in \mathcal{N}(A)$

$$\rightarrow \mathcal{L}(x, y) = f(\bar{x}) + \nabla_x f(\bar{x})^T \Delta x + \frac{1}{2} \Delta x^T \nabla_{xx}^2 f(\bar{x}) \Delta x + y^T (A \Delta x) = 0$$

$$\rightarrow \nabla_{\Delta x} \mathcal{L}(x, y) = \nabla_x f(\bar{x}) + \nabla_{xx}^2 f(\bar{x}) \Delta x + A^T y = 0$$

$$\nabla_y \mathcal{L}(x, y) = A \Delta x = 0$$

$$\Rightarrow \begin{pmatrix} \nabla_{xx}^2 f(\bar{x}) & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} \Delta x \\ y \end{pmatrix} = - \begin{pmatrix} \nabla_x f(\bar{x}) \\ 0 \end{pmatrix} \quad \begin{array}{l} \text{Feasible start} \\ \text{when } Ax^{(0)} = b \end{array}$$

Schur complement

If we want to invert:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} c \\ d \end{pmatrix}$$

$$A \in \mathbb{R}^{p \times p} \quad D \in \mathbb{R}^{q \times q}$$

$\rightarrow$ if A is invertible, define Schur complement

$$S = D - CA^{-1}B \in \mathbb{R}^{q \times q}$$

Can use this to solve:

$$y = S^{-1}(d - CA^{-1}c)$$

$$x = A^{-1}(c - By)$$

$\Rightarrow$ thus, Schur complement allows us to avoid
inverting $(p+q) \times (p+q)$ matrix (i.e., $\mathcal{O}((p+q)^3)$)

Instead, we can reduce this to complexity $\mathcal{O}(p^3)$ or $\mathcal{O}(q^3)$